

UNIVERSITY OF L'AQUILA

DISIM

Master Degree in Applied Data Science

TIME SERIES

WITH APPLICATIONS ON BIG DATA

Global Mean Sea Level (GMSL) Subpolar of North Atlantic

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Introduction

This project aims to analyze a time series and develop various ARIMA models to determine the best-performing one. The analysis focuses on data which records the Global Mean Sea Level (GMSL) Subpolar of North Atlantic from 1900 to 2010 with $T(=111)$.

The data used come from (<https://doi.org/10.1038/s41586-020-2591-3>).

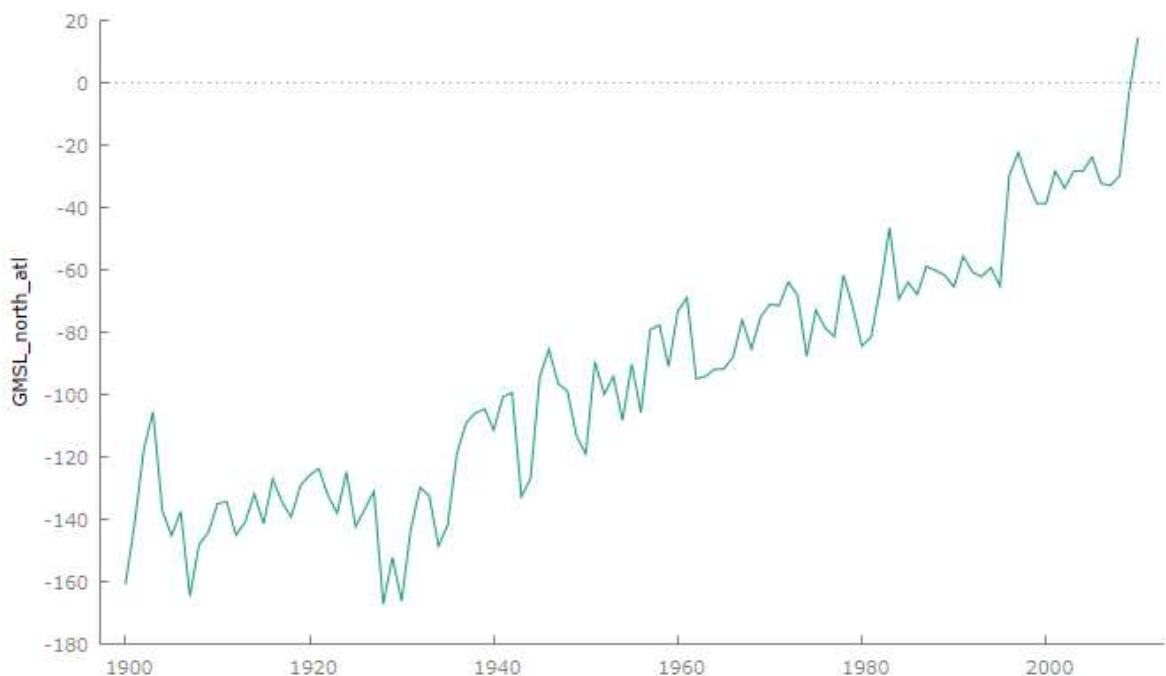
The process begins with testing the stationarity of the series. If the series is non-stationary, appropriate transformations are applied to make it stationary. The autocorrelation and partial autocorrelation functions (ACF and PACF) are then examined to understand the series' behavior. Using the Information Criteria method—AIC, BIC, and HQC—the most suitable models are identified.

The selected models are constructed and evaluated to ensure the residuals approximate a Gaussian white noise process. Finally, forecasts are generated, and the model with the lowest RMSE is chosen as the best model for predicting future values.

All analyses were conducted using GretL.

Preliminary Analysis

First, the series has been plotted and the result obtained is the following:



We can notice the presence of upward trend clearly from this figure, and in order to construct an ARIMA model, it is important to have a stationary series; therefore, I'm going to plot the first differenced time series, for making it more stationary:

Data Transformation

The first applied transformation is the first difference: $\Delta = 1 - L$

where L is the Lag Operator, and the new series obtained is $y_t = \Delta x_t = x_t - x_{t-1}$.

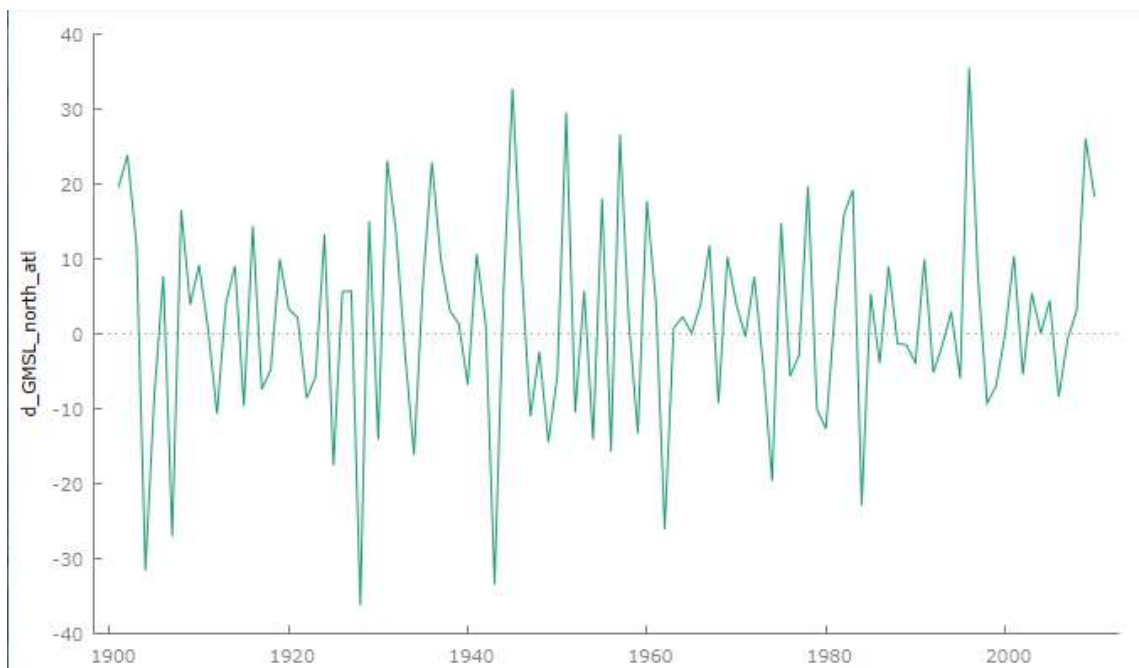
In order to check if the series is now stationary, a Python function which uses the Augmented Dickey-Fuller (ADF) stationarity test has been used.

```
test with constant
including 3 lags of (1-L)d_GMSL_north_atl
model: (1-L)y = b0 + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -2.17902
test statistic: tau_c(1) = -7.91197
asymptotic p-value 9.039e-13
1st-order autocorrelation coeff. for e: -0.035
lagged differences: F(3, 101) = 5.018 [0.0028]

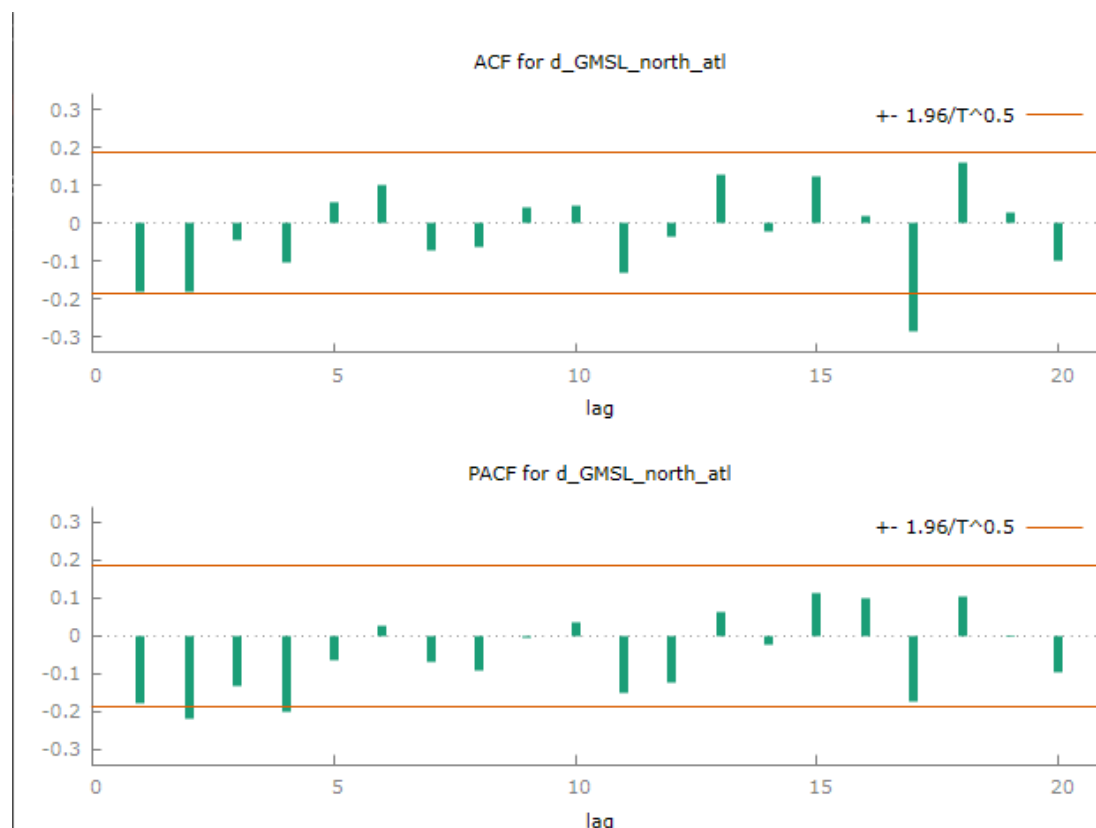
with constant and trend
including 3 lags of (1-L)d_GMSL_north_atl
model: (1-L)y = b0 + b1*t + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -2.23642
test statistic: tau_ct(1) = -8.10248
asymptotic p-value 8.877e-13
1st-order autocorrelation coeff. for e: -0.034
lagged differences: F(3, 100) = 5.479 [0.0016]
```

The ADF Test Statistic value is negative and suggests strong evidence for stationarity and the p-value is also low (close to zero), which suggests that one can reject the null hypothesis of non-stationarity; the p value must be less than a significance level of 0.05.

And the plot is the following:



In order to identify an ARMA(p,q) model, I'm going to examine the ACF, PACF and AIC, BIC, HQC criteria:



Interpretation:

The ACF and PACF plots suggest an AR(1) at lag 17 and MA(1) at lag 2, so let's proceed with the ARMA(1,1) model. Let's check now the criteria

Criteria for ARMA (p, q) specifications

Estimated using AS 197 (exact ML)
Dependent variable d_GMSL_north_atl, T = 88
Criteria for ARMA(p, q) specifications

p, q	AIC	BIC	HQC	loglik
0, 0	719.4226	724.3773	721.4188	-357.7113
0, 1	709.9524	717.3844	712.9465	-351.9762
0, 2	702.8794	712.7888	706.8717	-347.4397
0, 3	704.5411	716.9278	709.5314	-347.2705
1, 0	716.8865	724.3185	719.8806	-355.4432
1, 1	701.8240*	711.7334*	705.8162*	-346.9120
1, 2	703.7838	716.1705	708.7741	-346.8919
1, 3	704.9210	719.7850	710.9093	-346.4605
2, 0	712.8185	722.7279	716.8108	-352.4093
2, 1	703.7944	716.1810	708.7846	-346.8972
2, 2	705.4981	720.3621	711.4864	-346.7490
2, 3	706.9023	724.2436	713.8887	-346.4511
3, 0	712.2644	724.6511	717.2547	-351.1322
3, 1	705.3673	720.2313	711.3556	-346.6836
3, 2	707.2675	724.6088	714.2538	-346.6337
3, 3	706.5084	726.3271	714.4929	-345.2542

'*' indicates best, per criterion
Log-likelihood ('loglik') is provided for reference

Automatic criteria AIC, BIC, HQC

($0 \leq p \leq 3, 0 \leq q \leq 3$) suggest an ARMA (1,1) model.

** As both correlogram and criteria for ARMA predicted the same model, I decided to opte for the **second difference** in order to have at least two models to compare.

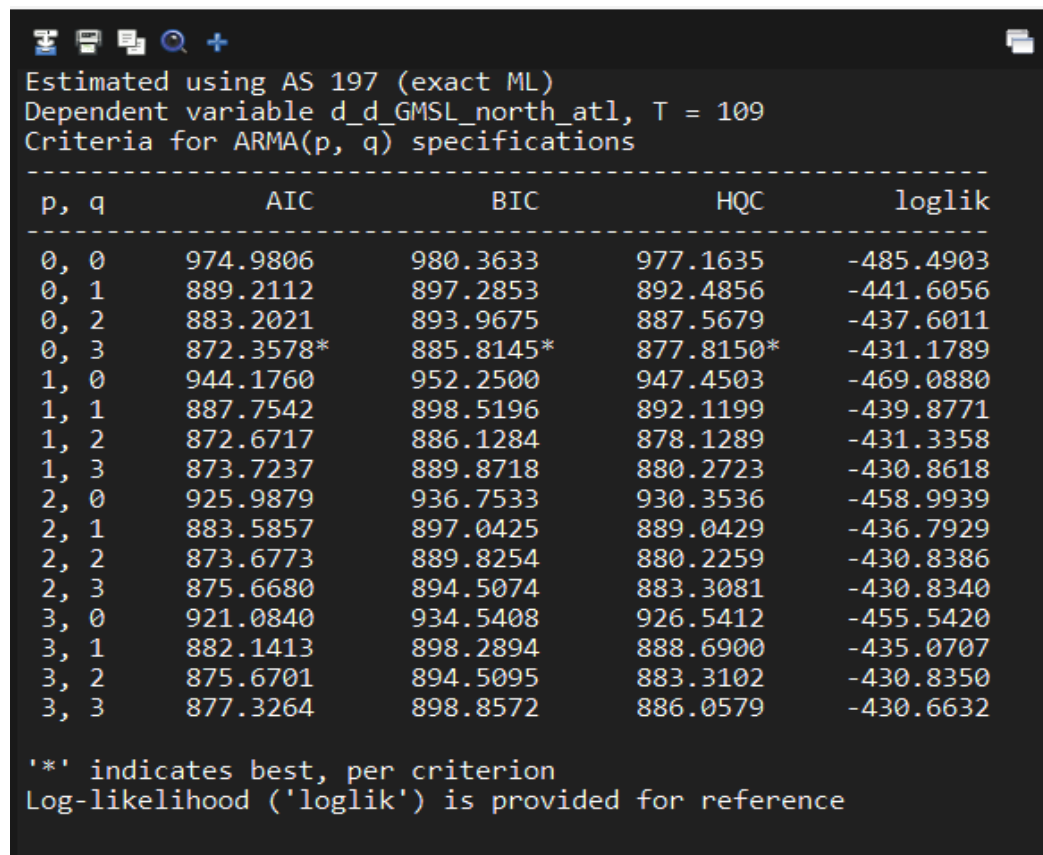
$\Delta^2 = (1-L)^2$ This means: $\Delta^2 x_t = x_t - 2x_{t-1} + x_{t-2}$



Interpretation:

The ACF and PACF plots suggest an AR(1) at lag 17 and MA(2) at lag 2 and 3, so I'll proceed with the ARMA(1,1) model. Let's check now the criteria

Criteria for ARMA(p, q) specifications



```
Estimated using AS 197 (exact ML)
Dependent variable d_d_GMSL_north_atl, T = 109
Criteria for ARMA(p, q) specifications
-----
p, q      AIC      BIC      HQC      loglik
-----
0, 0      974.9806   980.3633   977.1635   -485.4903
0, 1      889.2112   897.2853   892.4856   -441.6056
0, 2      883.2021   893.9675   887.5679   -437.6011
0, 3      872.3578*  885.8145*  877.8150*   -431.1789
1, 0      944.1760   952.2500   947.4503   -469.0880
1, 1      887.7542   898.5196   892.1199   -439.8771
1, 2      872.6717   886.1284   878.1289   -431.3358
1, 3      873.7237   889.8718   880.2723   -430.8618
2, 0      925.9879   936.7533   930.3536   -458.9939
2, 1      883.5857   897.0425   889.0429   -436.7929
2, 2      873.6773   889.8254   880.2259   -430.8386
2, 3      875.6680   894.5074   883.3081   -430.8340
3, 0      921.0840   934.5408   926.5412   -455.5420
3, 1      882.1413   898.2894   888.6900   -435.0707
3, 2      875.6701   894.5095   883.3102   -430.8350
3, 3      877.3264   898.8572   886.0579   -430.6632

'* indicates best, per criterion
Log-likelihood ('loglik') is provided for reference
```

Automatic criteria AIC, BIC, HQC

$(0 \leq p \leq 3, 0 \leq q \leq 3)$ suggest an ARMA (0,3) model.

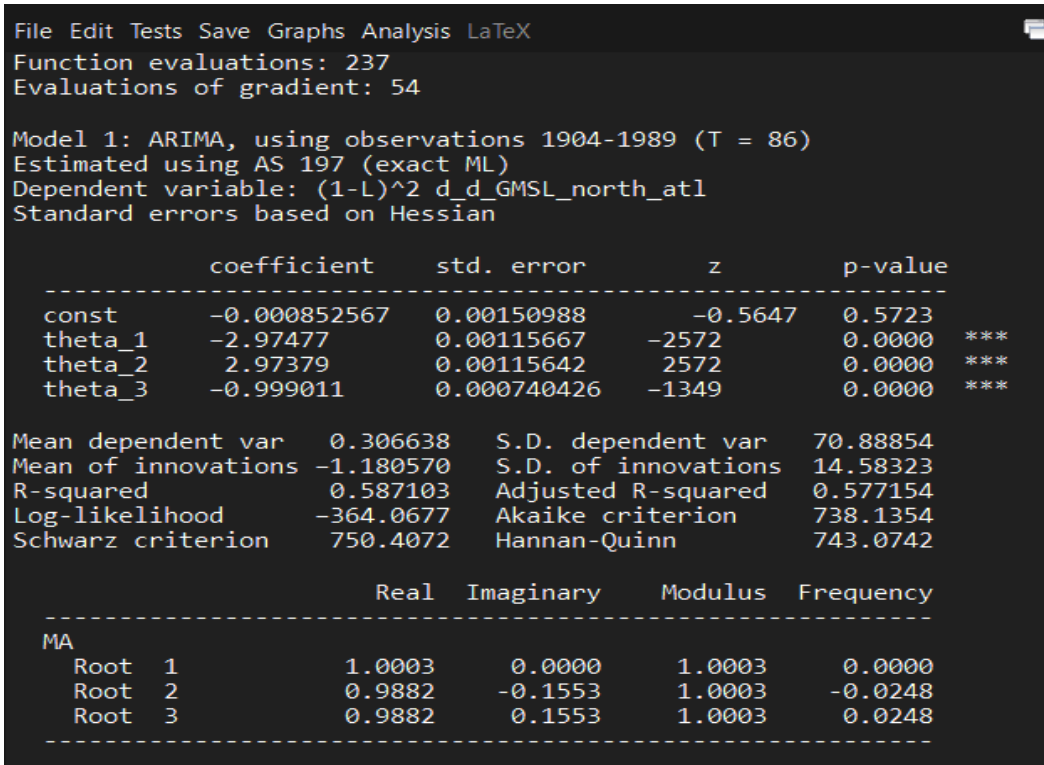
Model Estimation

Before the creation of the model, the time series has been divided into training set (80%) and test set (20%);

- Training Set: from 1900 to 1988
- Test Set: from 1989 to 2010

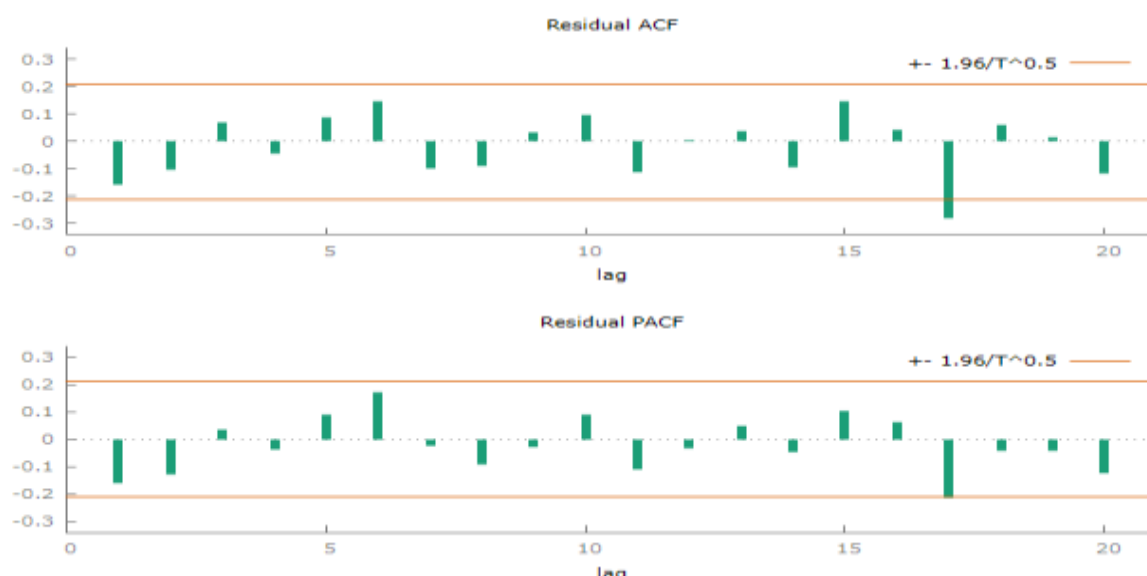
1) First model: ARIMA(0,2,3)

The estimation of this model has the following results:



Model Checking

Let's see the plot of residuals versus time

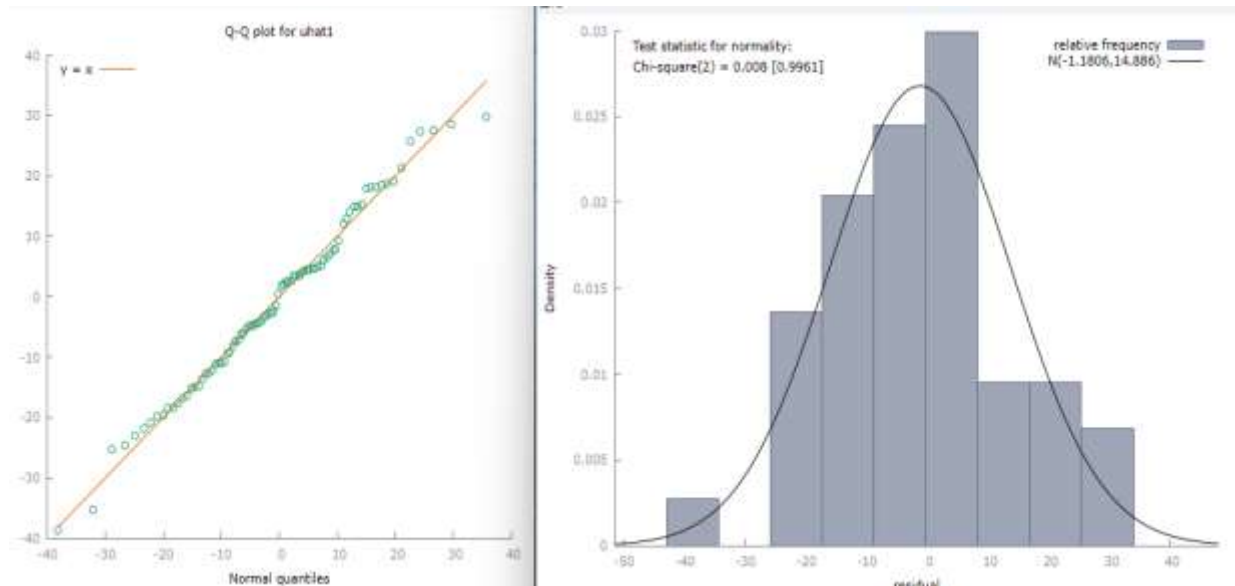


- Looking at the Residual ACF, for the first 20 lags, all values fall inside the 95% confidence bounds except one at lag 17 in ACF (not a relevant one), hence the residuals can be considered a realization of an iid process.

- The residuals appear to be white noise, meaning they are uncorrelated and random.

- This is a good sign that the ARMA(0,3) model is appropriate, as it has adequately explained the underlying structure of the time series

let's now see if they are a realization of a Gaussian White Noise by analyzing the normality and the Q-Q plot.



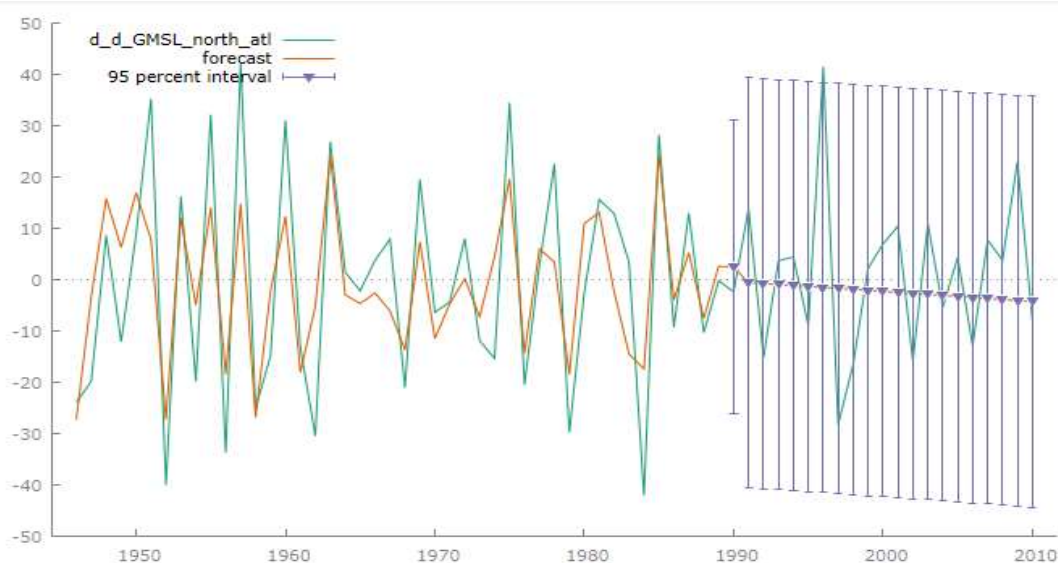
Moreover, looking at the Q-Q plot for the residuals, almost all data points fall along the straight line, hence here it is another evidence that the residuals are normally distributed.

From Chi-square (2) = 0.008 it is possible to say that the residuals follow a Gaussian distribution (in fact p-value=0.9961 $\gg$ 0.05);

Moreover, looking at the Q-Q plot for the residuals, almost all data points fall along the straight line, hence here it is another evidence that the residuals are normally distributed.

Forecast

By doing the forecast on the test set, the following result is obtained



1985	28.211089	24.265638			
1986	-9.274754	-3.930194			
1987	12.943856	5.252502			
1988	-10.323977	-7.569144			
1989	-0.191552	2.614846			
1990	-2.362899	2.412374	14.583228	-26.170229	-30.994976
1991	13.849530	-0.578084	20.365311	-40.493361	-39.337192
1992	-15.142026	-0.765773	20.368381	-40.687065	-39.155520
1993	3.653279	-0.954313	20.371451	-40.881624	-38.972998
1994	4.433892	-1.143706	20.374523	-41.077039	-38.789626
1995	-8.895678	-1.333952	20.377597	-41.273308	-38.605404
1996	41.462860	-1.525051	20.380672	-41.470433	-38.420332
1997	-28.213884	-1.717002	20.383748	-41.668414	-38.234411
1998	-16.531873	-1.909805	20.386826	-41.867249	-38.047639
1999	2.221320	-2.103461	20.389905	-42.066940	-37.860018
2000	6.955182	-2.297970	20.392985	-42.267486	-37.671547
2001	10.453685	-2.493331	20.396067	-42.468888	-37.482226
2002	-15.843487	-2.689545	20.399150	-42.671144	-37.292055
2003	10.878574	-2.886611	20.402235	-42.874256	-37.101034
2004	-5.350870	-3.084530	20.405321	-43.078224	-36.909164
2005	4.358319	-3.283301	20.408408	-43.283046	-36.716444
2006	-12.832761	-3.482925	20.411497	-43.488724	-36.522873
2007	7.724802	-3.683402	20.414587	-43.695257	-36.328453
2008	3.802931	-3.884731	20.417679	-43.902646	-36.133184
2009	22.978177	-4.086913	20.420772	-44.110890	-35.937064
2010	-7.862904	-4.289947	20.423866	-44.319989	-35.740094

Forecast evaluation statistics using 21 observations	
Mean Error	3.1199
Root Mean Squared Error	15.478
Mean Absolute Error	12.279
Mean Percentage Error	118.52
Mean Absolute Percentage Error	118.52
Theil's U2	0.95808
Bias proportion, UM	0.040631
Regression proportion, UR	0.025801
Disturbance proportion, UD	0.93357

- The model demonstrates modest accuracy, as indicated by RMSE and MAE, but struggles to provide precise predictions (evident from high MAPE and wide confidence intervals).
- Random noise (disturbances) constitutes the primary source of error, limiting the model's ability to capture short-term fluctuations in the series.
- Despite these limitations, the model may still be suitable for understanding general trends and producing rough forecasts, though improvements (e.g., incorporating additional variables or refining the model) may enhance performance.

2) First model: ARIMA (1,2,2)

The estimation of this model has the following results:

```

File Edit Tests Save Graphs Analysis LaTeX
Function evaluations: 130
Evaluations of gradient: 34

Model 3: ARIMA, using observations 1904-1989 (T = 86)
Estimated using AS 197 (exact ML)
Dependent variable: (1-L)^2 d_d_GMSL_north_atl
Standard errors based on Outer Products matrix
Warning: couldn't compute numerical Hessian

```

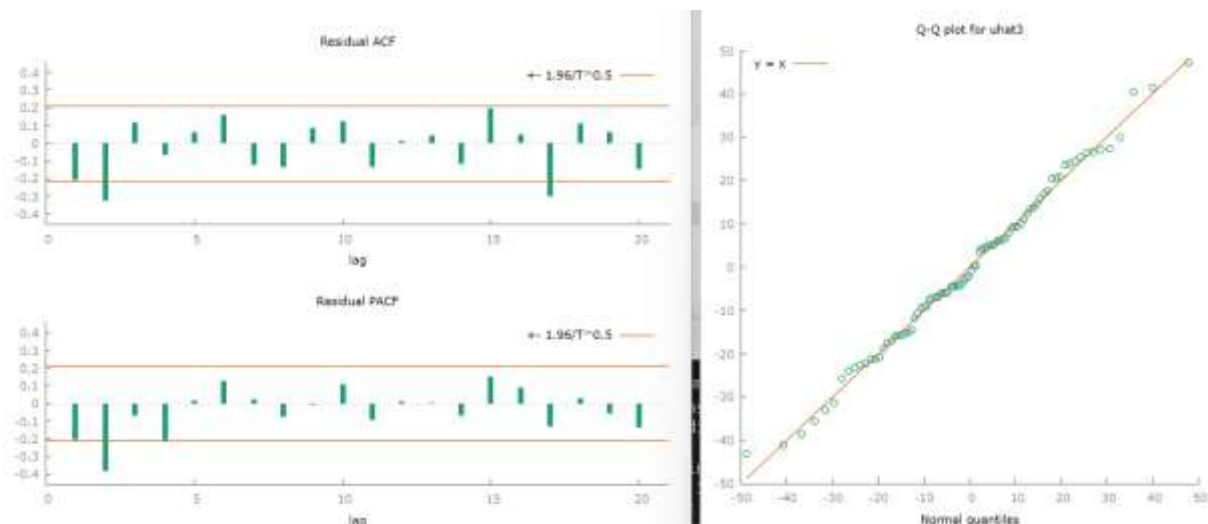
	coefficient	std. error	z	p-value
const	-0.00375684	0.0170288	-0.2206	0.8254
phi_1	-0.502964	0.0929533	-5.411	6.27e-08 ***
theta_1	-1.98803	0.214417	-9.272	1.83e-20 ***
theta_2	1.00000	0.216026	4.629	3.67e-06 ***

Mean dependent var	0.306638	S.D. dependent var	70.88854
Mean of innovations	-0.436946	S.D. of innovations	19.07058
R-squared	0.261230	Adjusted R-squared	0.243428
Log-likelihood	-382.5178	Akaike criterion	775.0356
Schwarz criterion	787.3074	Hannan-Quinn	779.9744

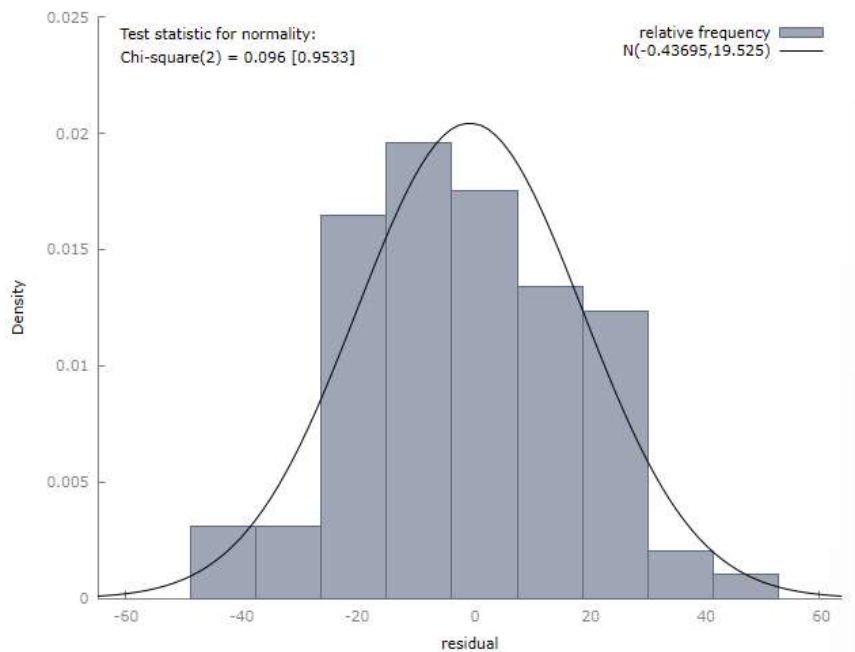
	Real	Imaginary	Modulus	Frequency
AR				
Root 1	-1.9882	0.0000	1.9882	0.5000
MA				
Root 1	0.9940	-0.1093	1.0000	-0.0174
Root 2	0.9940	0.1093	1.0000	0.0174

Model Checking

Let's see the plot of residuals versus time



- The residuals appear to be white noise, meaning they are uncorrelated and random.
- Looking at the Residuals, for the first 20 lags, all values fall inside the 95% confidence bounds except one at lag 2, hence the residuals can be considered a realization of an iid process.
- This is a good sign that the ARMA (0,3) model is appropriate, as it has adequately explained the underlying structure of the time series



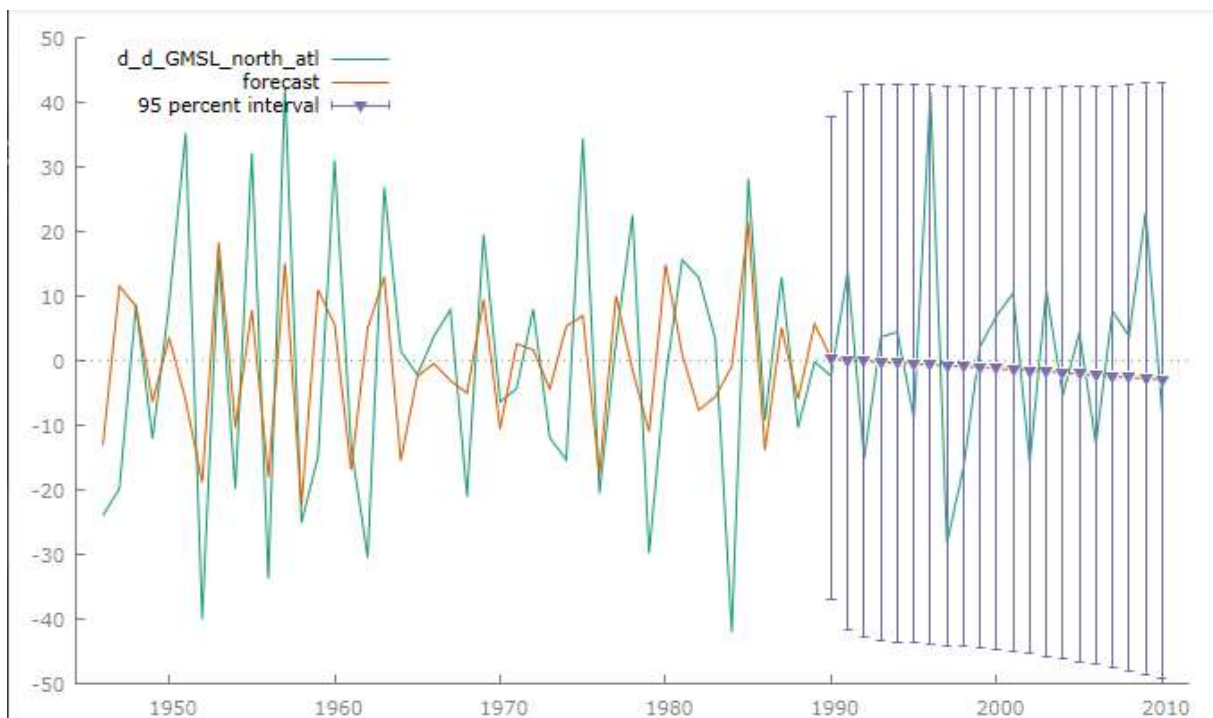
Moreover, looking at the Q-Q plot for the residuals, almost all data points fall along the straight line, hence here it is another evidence that the residuals are normally distributed.

From Chi-square(2) = 0.096 it is possible to say that the residuals follow a Gaussian distribution (in fact p-value=0.95331 $\gg$ 0.05);

Moreover, looking at the Q-Q plot for the residuals, almost all data points fall along the straight line, hence here it is another evidence that the residuals are normally distributed.

Forecast

By doing the forecast on the test set, the following result is obtained



Forecast evaluation statistics using 21 observations

Mean Error	2.2177
Root Mean Squared Error	15.251
Mean Absolute Error	11.994
Mean Percentage Error	108.2
Mean Absolute Percentage Error	108.2
Theil's U2	0.91892
Bias proportion, UM	0.021146
Regression proportion, UR	0.018181
Disturbance proportion, UD	0.96067

3) Selection of the best model

** Comparison to the Previous Model:

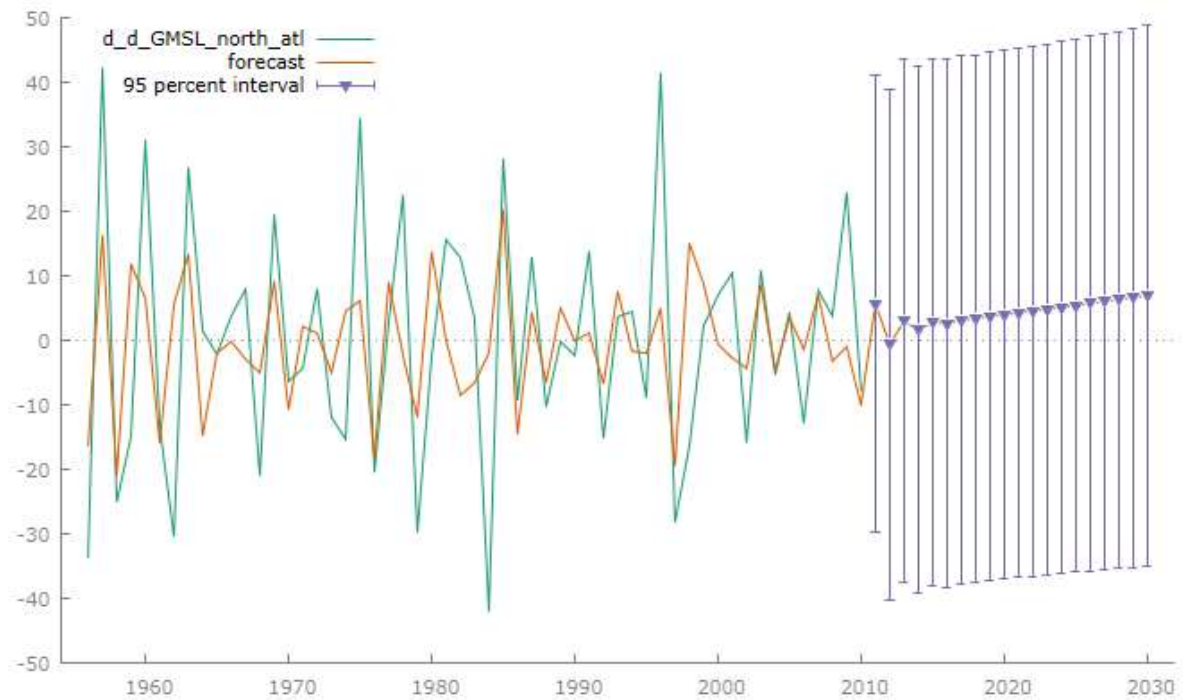
- **Improved Bias:** The Mean Error has decreased compared to the earlier results, indicating better alignment between the forecasted and actual values.
- **Smaller Proportions of Bias and Regression Errors:** Suggesting better model fitting and reduction in systematic errors.
- **High Noise Contribution (UD):** However, random disturbances remain the dominant source of errors, consistent with inherent noise in the data.

In our case, comparing the two root mean square errors, we have that the best model is the **ARIMA(1, 2, 2)**.

4) Forecasts of our time series

Finally, after finding out the best model, we can use it to predict the future values of the rain millimeters up to the year 2030:

2010	-7.862904	-10.102442			
2011	5.697377	18.103064	-29.783975	-41.178730	
2012	-0.608916	20.160370	-40.122515	-38.904683	
2013	2.974955	20.700483	-37.597246	-43.547156	
2014	1.634376	20.789143	-39.111596	-42.380348	
2015	2.744914	20.844349	-38.109259	-43.599087	
2016	2.634587	20.844417	-38.219719	-43.488894	
2017	3.131519	20.862494	-37.758217	-44.021256	
2018	3.325566	20.869736	-37.578365	-44.229497	
2019	3.669849	20.886983	-37.267885	-44.607582	
2020	3.938777	20.903871	-37.032057	-44.909612	
2021	4.244664	20.927048	-36.771586	-45.260925	
2022	4.531594	20.953522	-36.536554	-45.599742	
2023	4.827404	20.985564	-36.303546	-45.958353	
2024	5.118235	21.022558	-36.085221	-46.321690	
2025	5.410987	21.065437	-35.876511	-46.698484	
2026	5.702225	21.114291	-35.681024	-47.085474	
2027	5.993650	21.169646	-35.498085	-47.485404	
2028	6.284438	21.231785	-35.329095	-47.897972	
2029	6.574986	21.301101	-35.174405	-48.324377	
2030	6.865092	21.377914	-35.034849	-48.765032	



Conclusion

The forecasted values trend around a stable mean, with increasing uncertainty as time progresses (widening intervals), which is typical of time-series forecasts.

The model indicates that the future values of the series are expected to remain within historical variability limits, with no dramatic trend upward or downward.

The ARIMA model provides a reliable short-term forecast of the time series with interpretable results and reasonable uncertainty bounds. While it captures the patterns in the data well, long-term projections should be approached cautiously, considering potential external factors. The insights derived from this project can guide further research or policy recommendations based on the series' expected behavior.